Extra Credit

Orientation and Course Materials

- Welcome & Overview
- How to navigate the SOC 170 Canvas Environment
- How to Get Help | Office Hours, IT, and Student Wellness Support
- What do I do if I get sick or need to quarantine?
- 170_Syllabus_Spring26_v3.docx
- Soc170_[Dis 301,302,303,304]_Syllabus.pdf
- Extra Credit

Extra Credit

SUBMIT HERE

Extra Credit

Attend one of the following Seminars, take a picture, and write <500 words that (a) summarize the talk and (b) describe something new that you learned.
Submit both together AS A SINGLE PDF here.

ALL OF THESE ARE SUBJECT TO CHANGE - CHECK THE DATE/TIME/LOCATION BEFORE YOU GO!

Here are the website of the speaker series.

https://pophealth.wisc.edu/events/

https://www.irp.wisc.edu/2026-spring-irp-seminar-calendar/

https://cde.wisc.edu/events/demsem/

Thursday, January 22, 2026
12:15-1:30 PM
Speaker: Daniel Galvin, Professor, Department of Political Science and Institute for Policy Research, Northwestern University
Title: Powers and Practices in Labor Standards Enforcement
Location: 8417 Sewell Social Science Building
Series: IRP Seminar

Thursday, January 29, 2026
12:15-1:30 PM
Speaker: Silvia Helena Barcellos, Associate Professor, Population Health Sciences and La Follette School of Public Affairs, University of Wisconsin-Madison
Title: A Chip Off The Old Block? Genetics and The Intergenerational Transmission of Socioeconomic Status
Location: 8417 Sewell Social Science Building
Series: IRP Seminar

Does Population Growth Threaten the Environment?

Lecture 1.6

2/5/26

I = PAT

- I = Human **impact** on the environment
- P = Population size
- A = Affluence (per capita consumption)
- T = Technology

It is more accurate to say that consumers, rather than people, cause climate change. (Stephenson et al. 2010)

- (Ehrlich & Holdren 1971)

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Ehrlich focused on P + A

Simon would have focused on T

t is more accurate to say that
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(Stephenson et al. 2010)

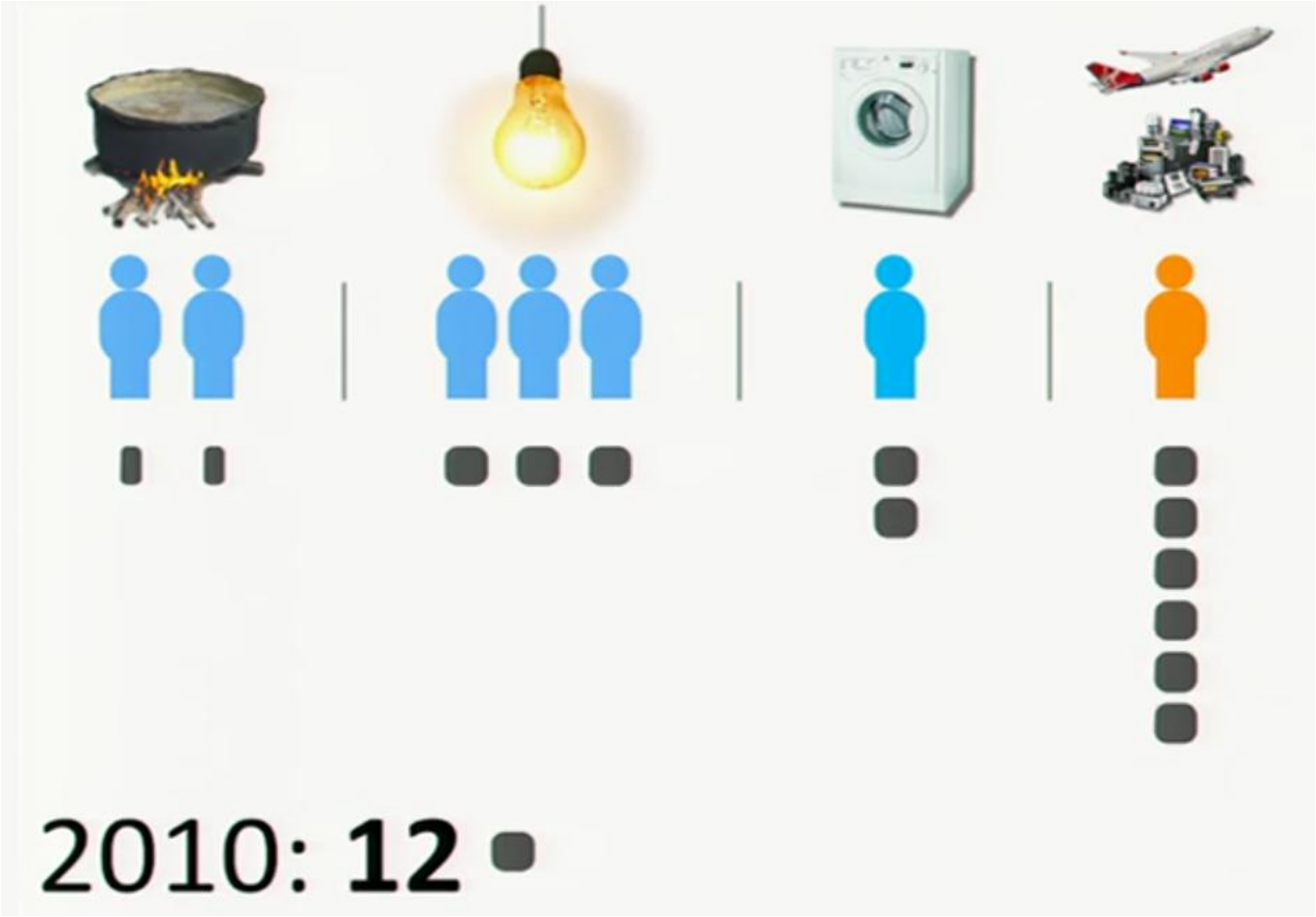
- (Ehrlich & Holdren 1971)

Why does **affluence** matter?

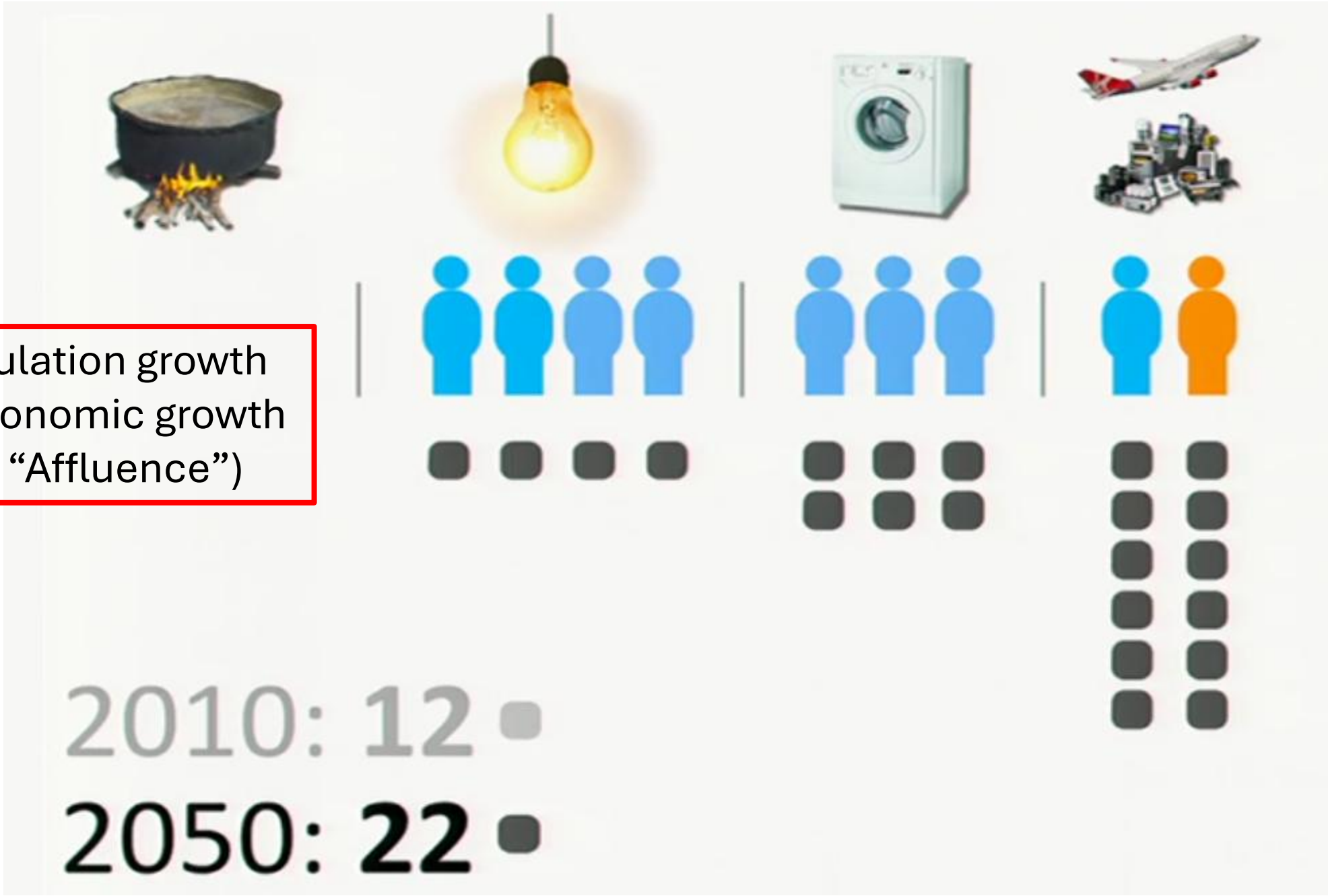
- As the **consumption** of individuals increases, the total environmental impact increases as well
  - Resources used to produce a growing number of goods
  - Waste generated in creating, transporting, and disposing of goods

Affluence and Consumption

- **Hans Rosling**
- As people get richer, they want to consume more, replacing manual labor


2010: 12

Population growth
+ economic growth
(aka “Affluence”)



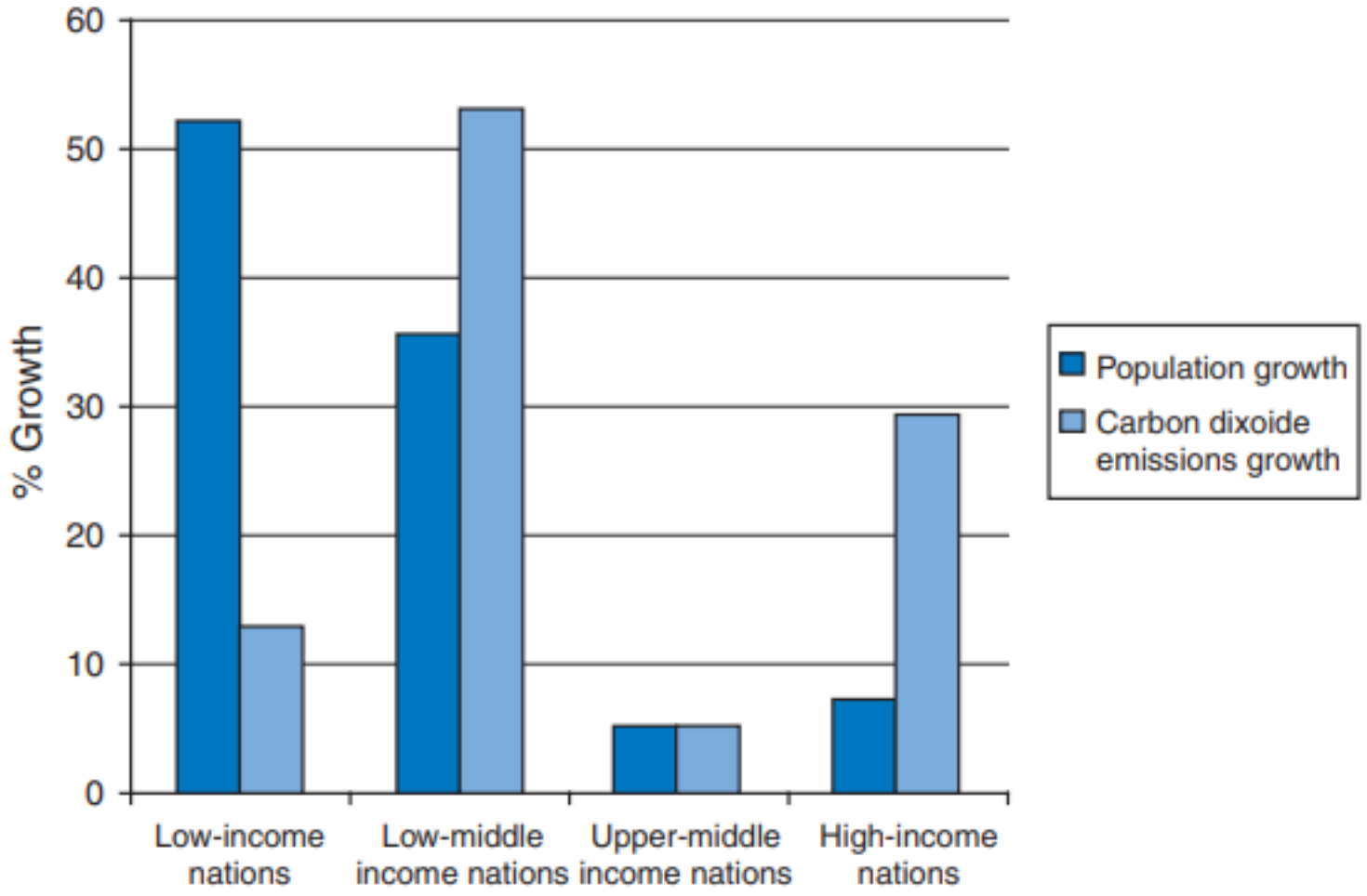


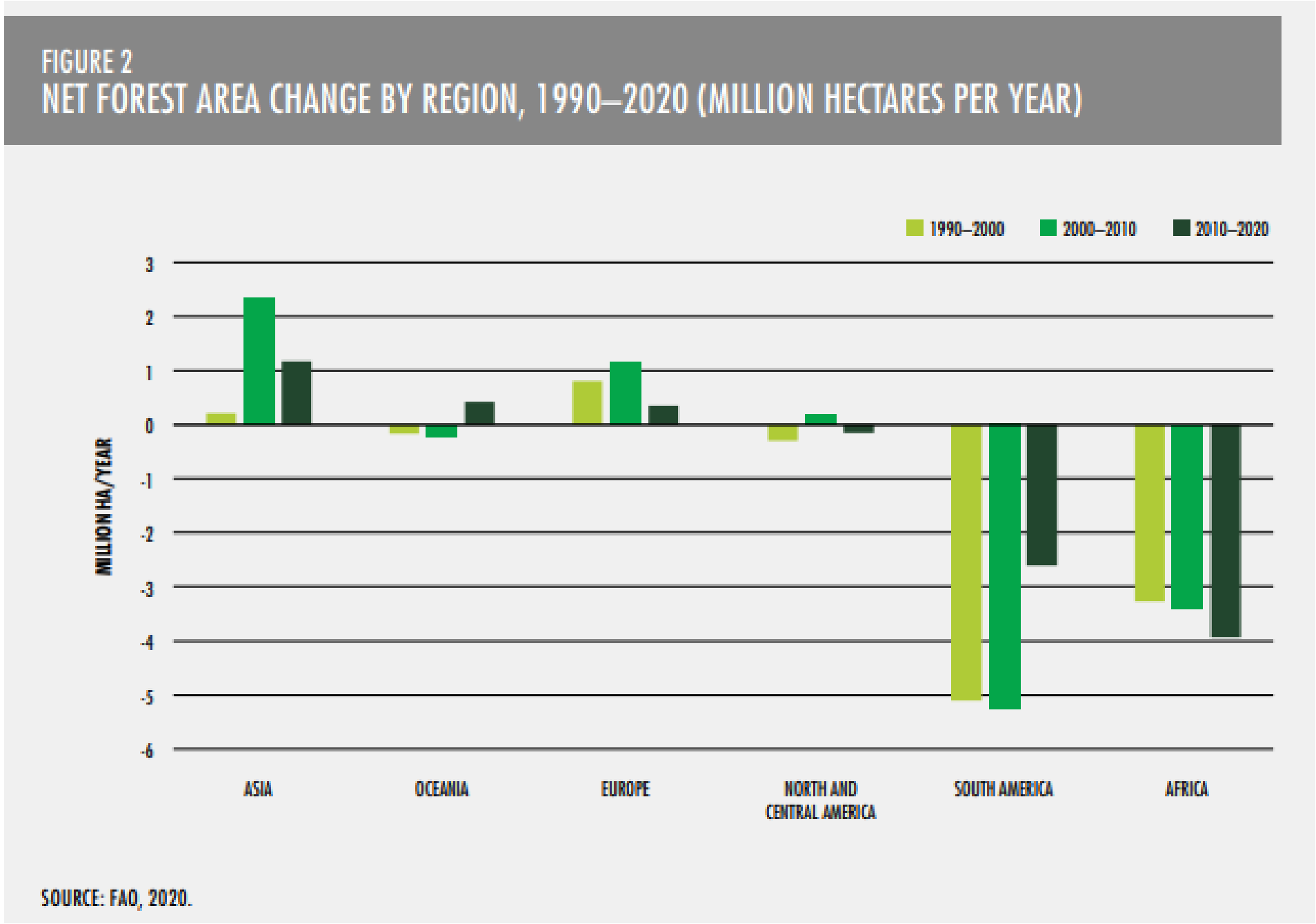
Fig. 3 Contribution to the growth in world population and CO_2 emissions by groups of nations classified according to their average per capita income levels, 1980–2005. Source: Satterthwaite, 2009.[14]

Source: Stephenson, Newman, and Mayhew (2010)

Why is it hard to disentangle population growth and environmental impact?

- Role of “affluence”
  - Influences both population growth and environmental impact
    - More affluent have fewer children!
  - The impact of poverty vs. the impact of wealth
    - Poverty: environmental vulnerability
    - Wealth: adaptation and consumption

Is Deforestation a *Population* Problem?



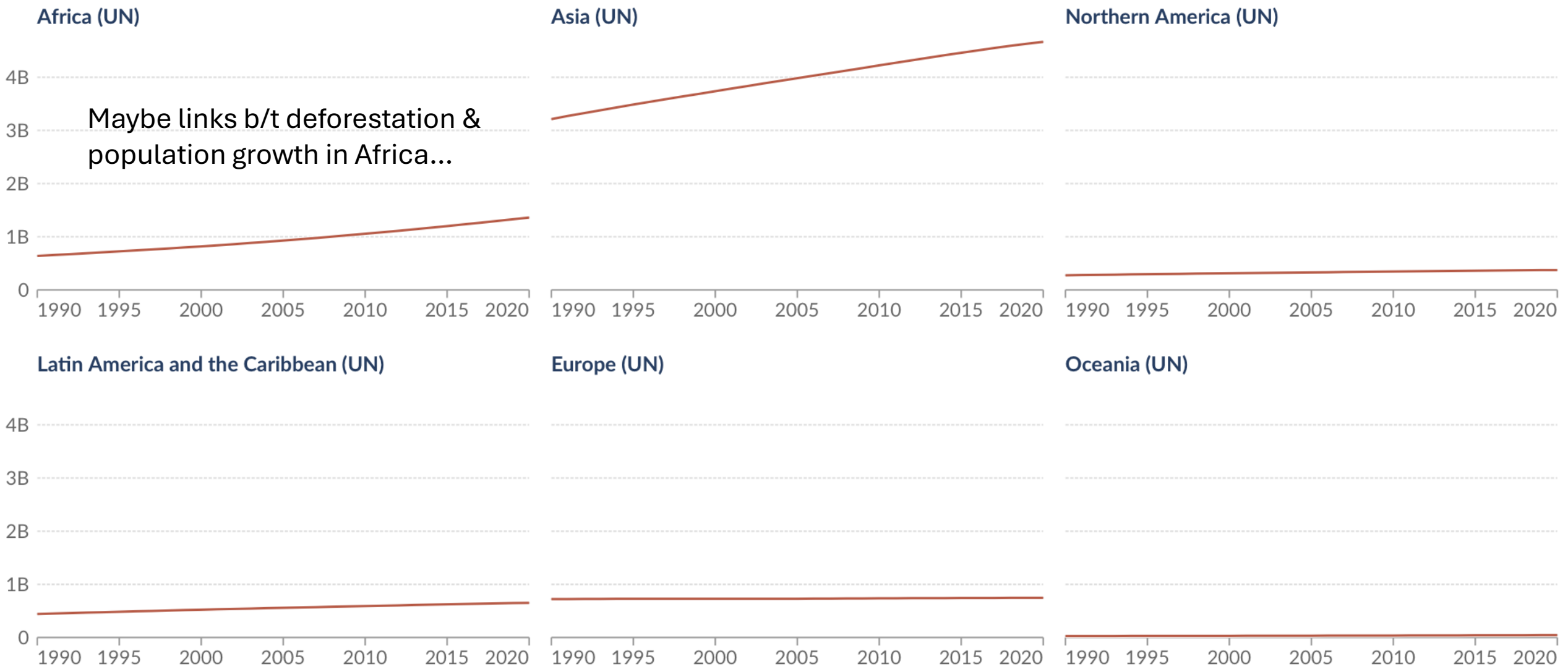
Source: FAO 2020

Population, 1990 to 2020

Our World in Data

Split by country or region

Align axis scales



Maybe links b/t deforestation & population growth in Africa...

Source: United Nations, World Population Prospects (2022)

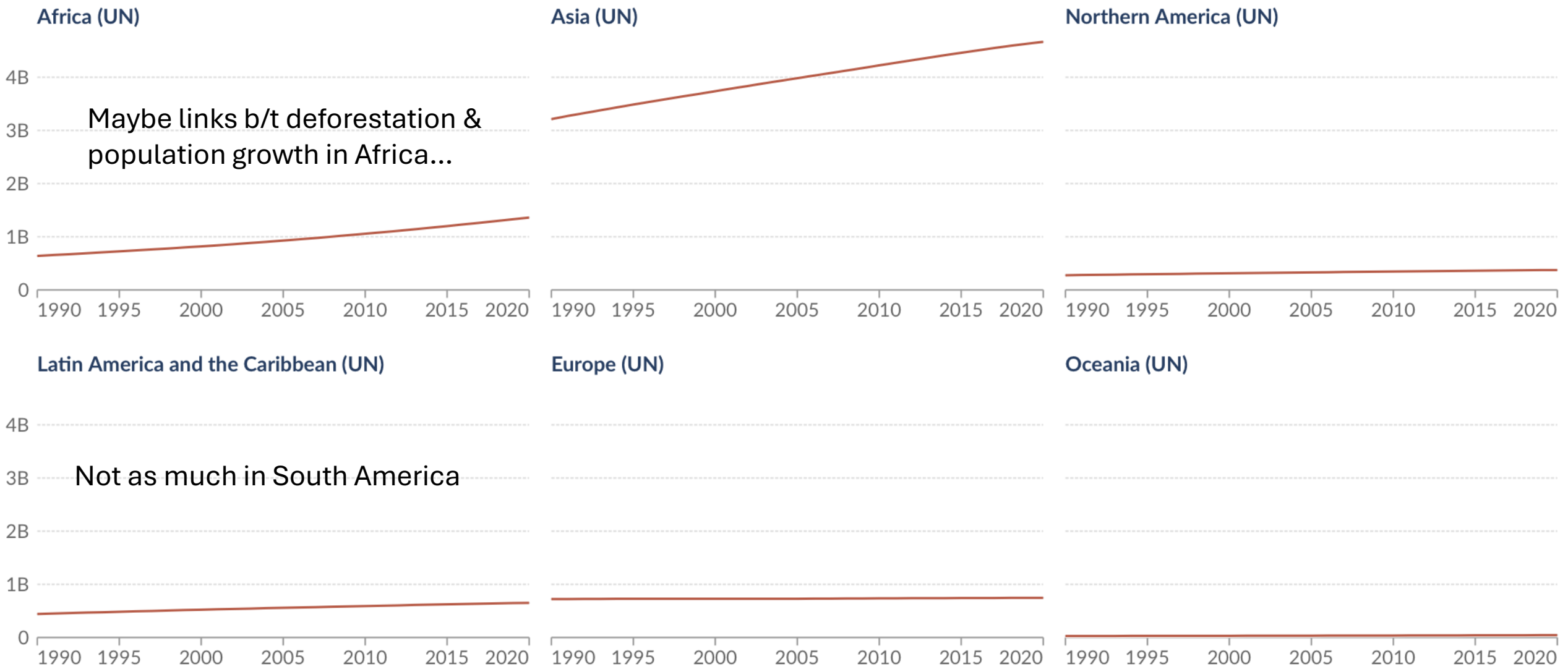
CC BY

Population, 1990 to 2020

Our World in Data

Split by country or region

☑ Align axis scales



Source: United Nations, World Population Prospects (2022)

CC BY

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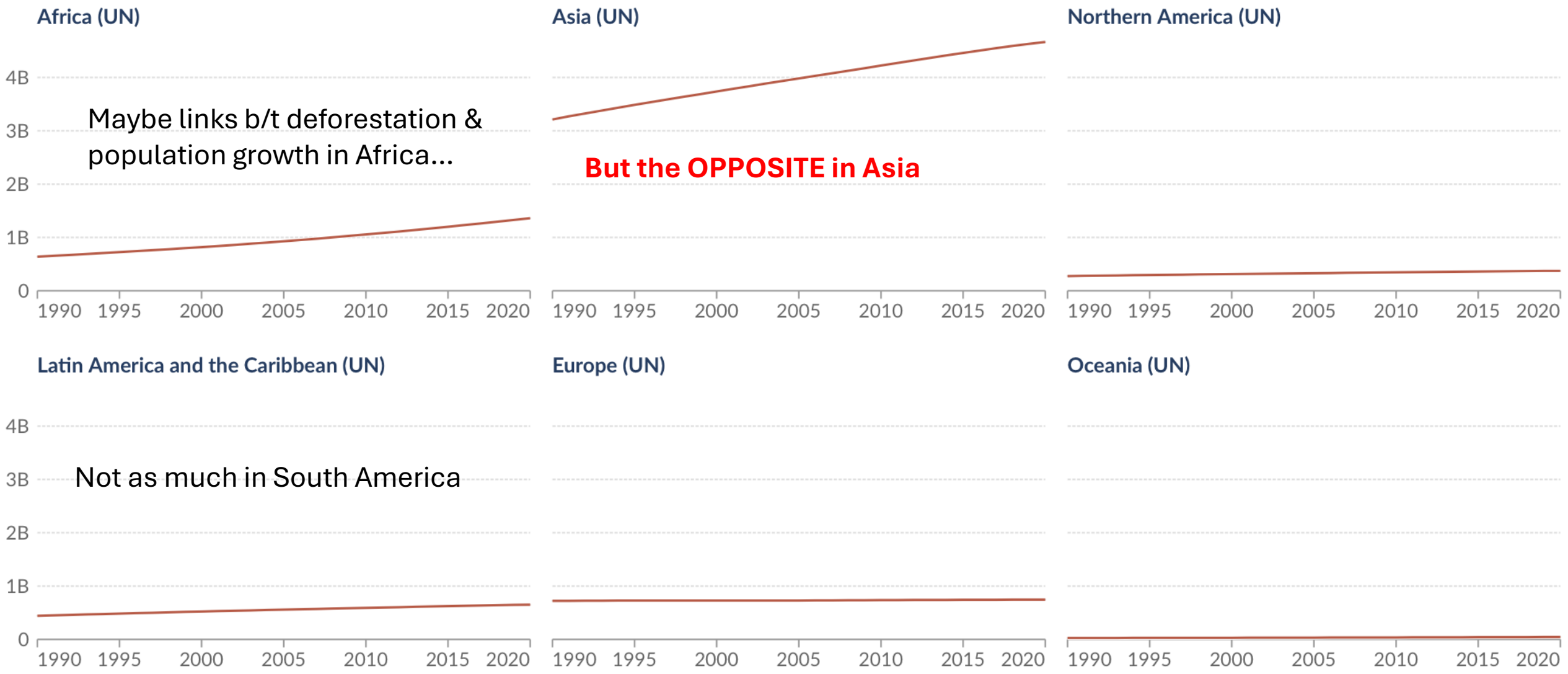

Source: United Nations, World Population Prospects (2022)

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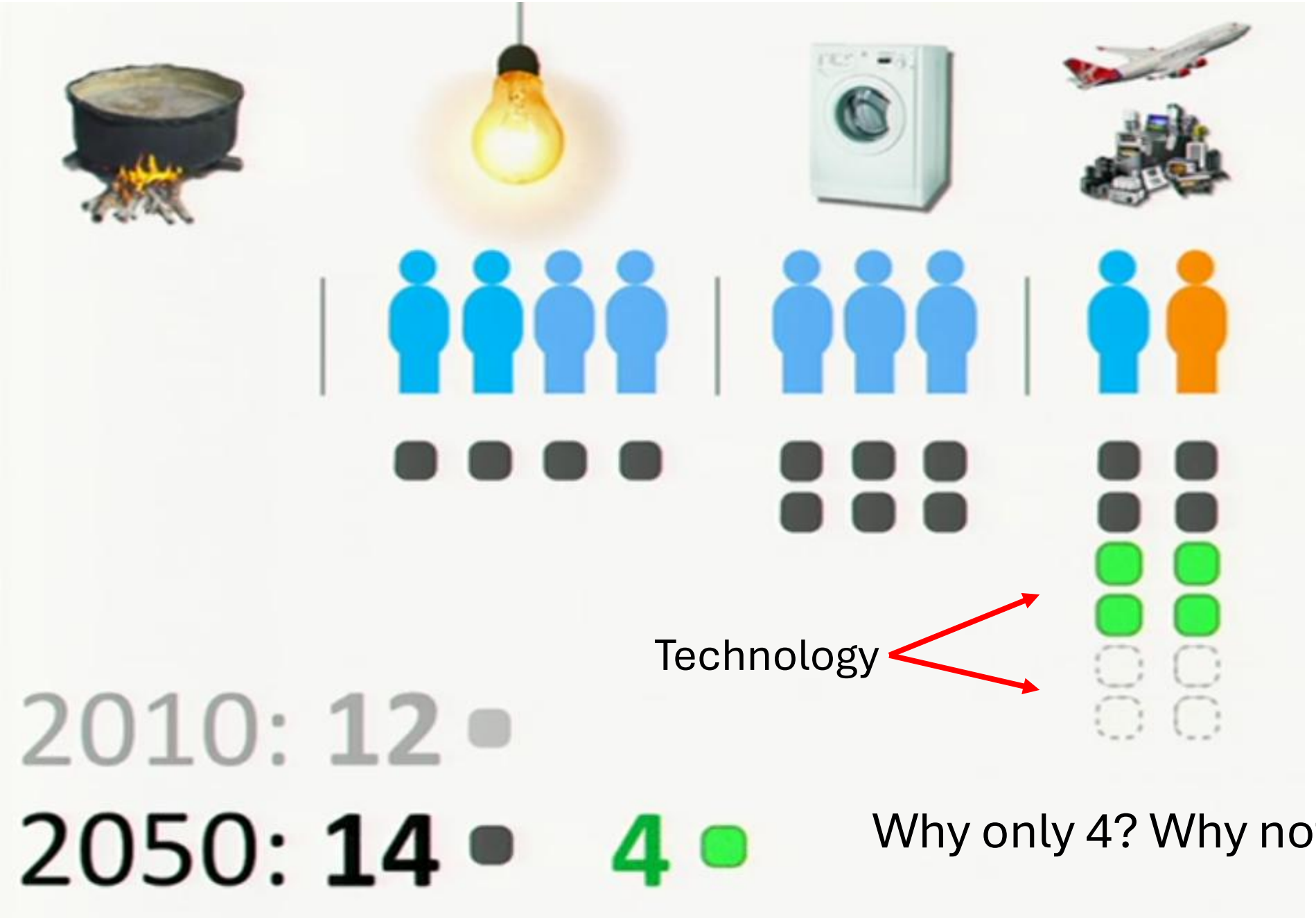
Technology

- Technology is used to produce more goods, but it also makes the production and disposal of goods more efficient

- Includes modes of production, fuel sources, waste/recycling/ carbon capture



Why only 4? Why not more?

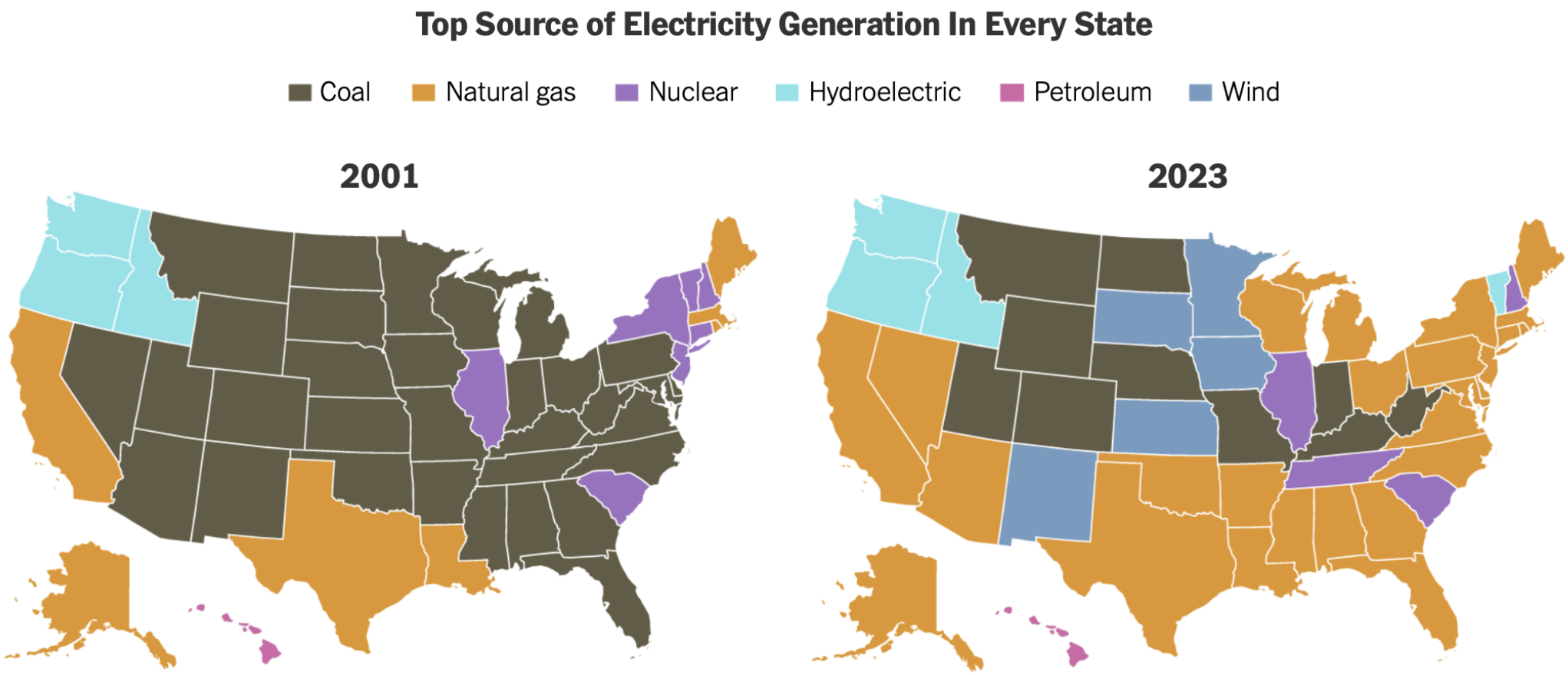
Technology and optimism

- **Hans Rosling**
- Is protecting the environment necessarily counter to the interest of development?

Environmental economics

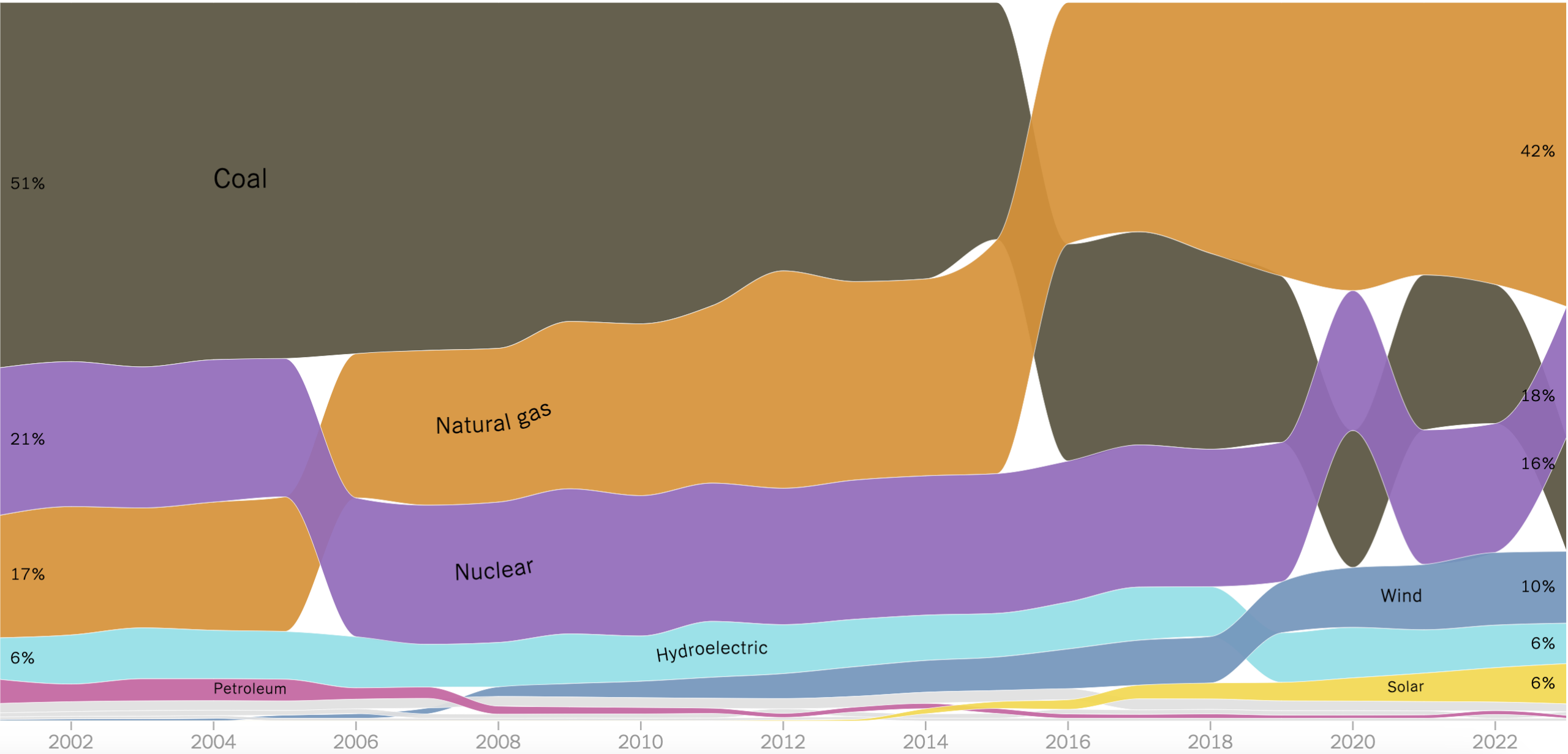
- Scare resources influence prices, influence demand
- Concern for resource availability drives knowledge production

Sources of Power are Changing Quickly



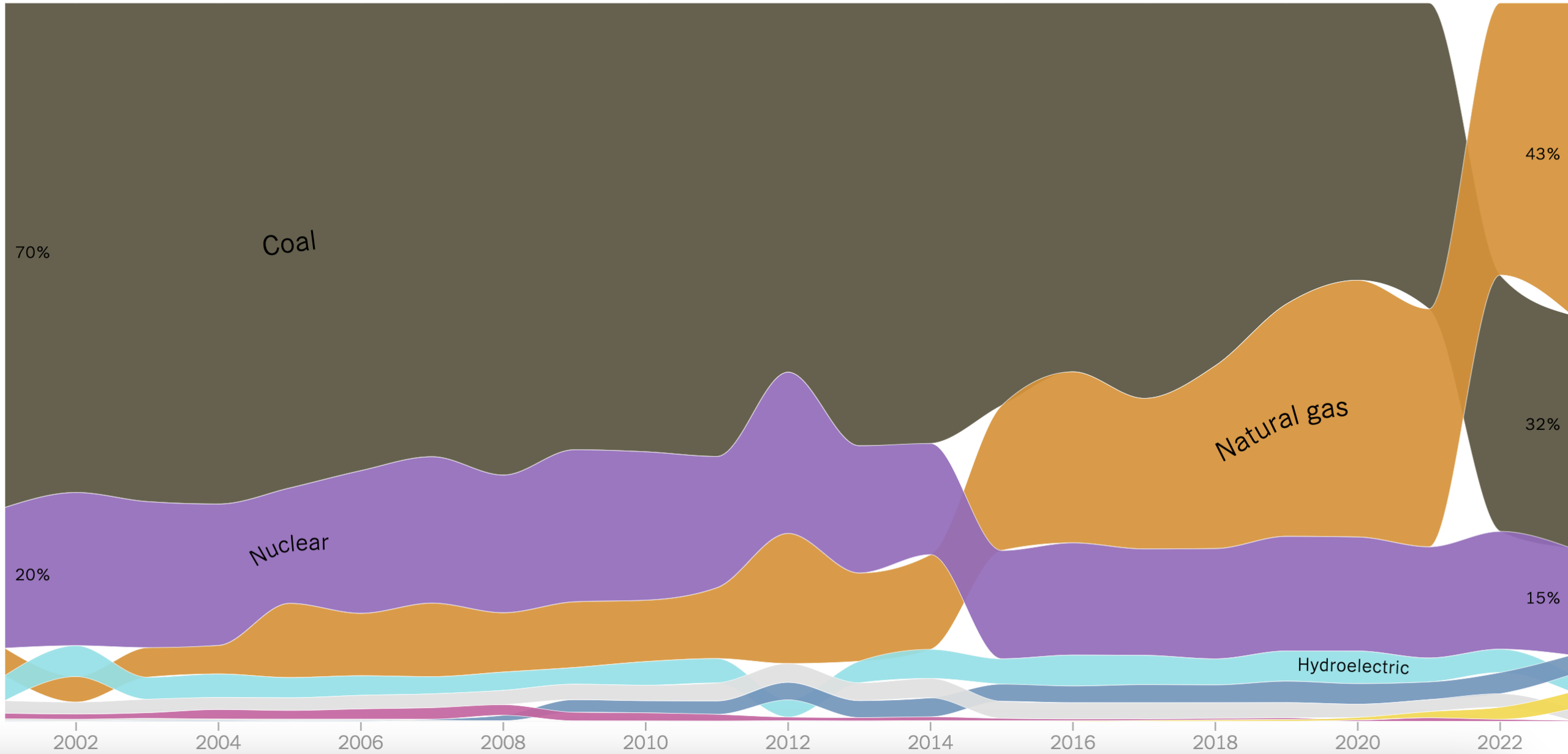
Source: U.S. Energy Information Administration

USA


51%
Coal
42%
Natural gas
21%
18%
16%
Nuclear
17%
Wind
10%
6%
Hydroelectric
6%
Petroleum
Solar
6%
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022

How **Wisconsin** made electricity from 2001 to 2023

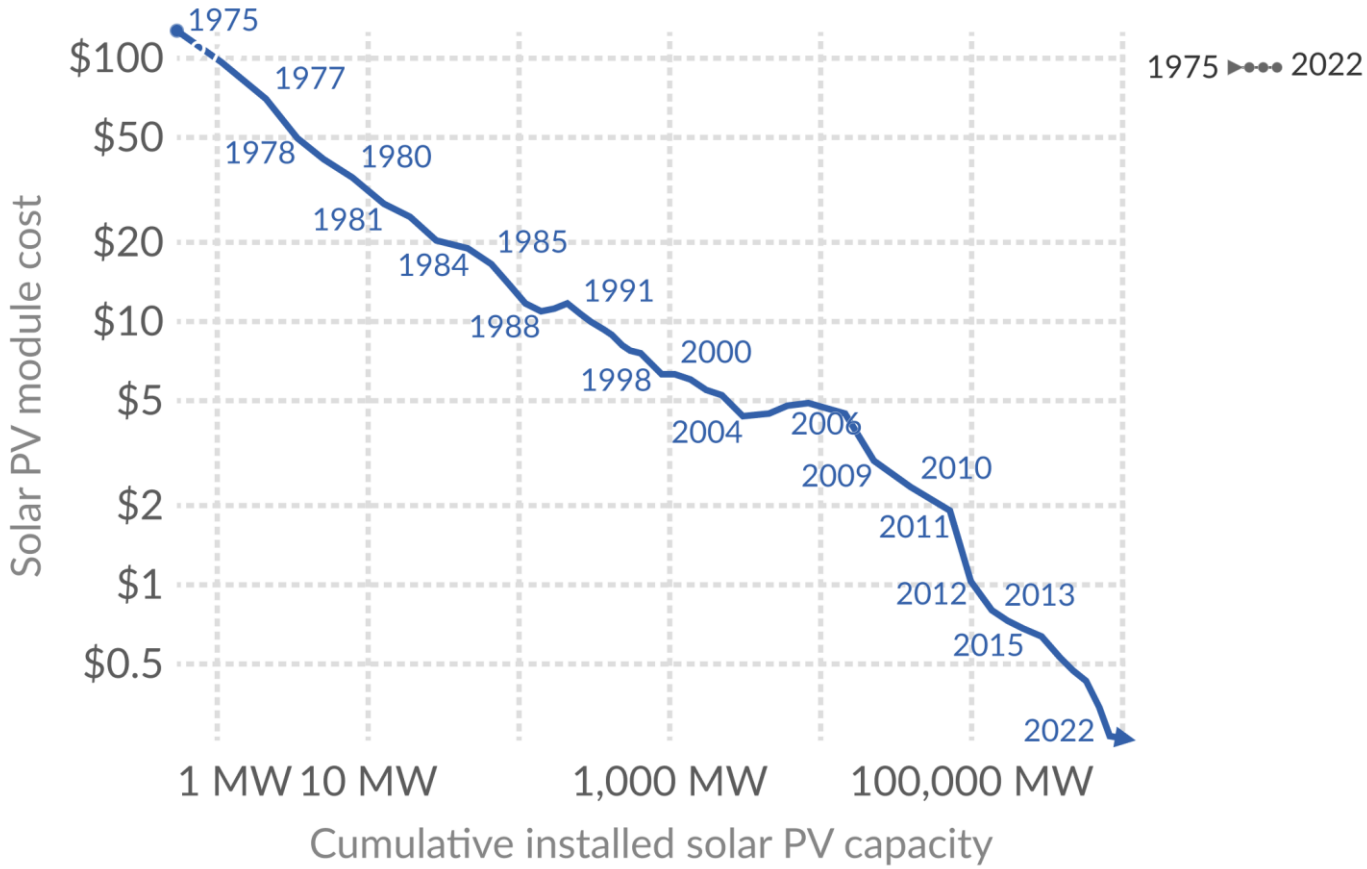
Percentage of power produced from each energy source



Solar panel costs have fallen by around 20% for every doubling of global cumulative capacity

Our World in Data

Costs are measured in US dollars per Watt, adjusted for inflation.



Data source: IRENA (2023); Nemet (2009); Farmer and Lafond (2016)

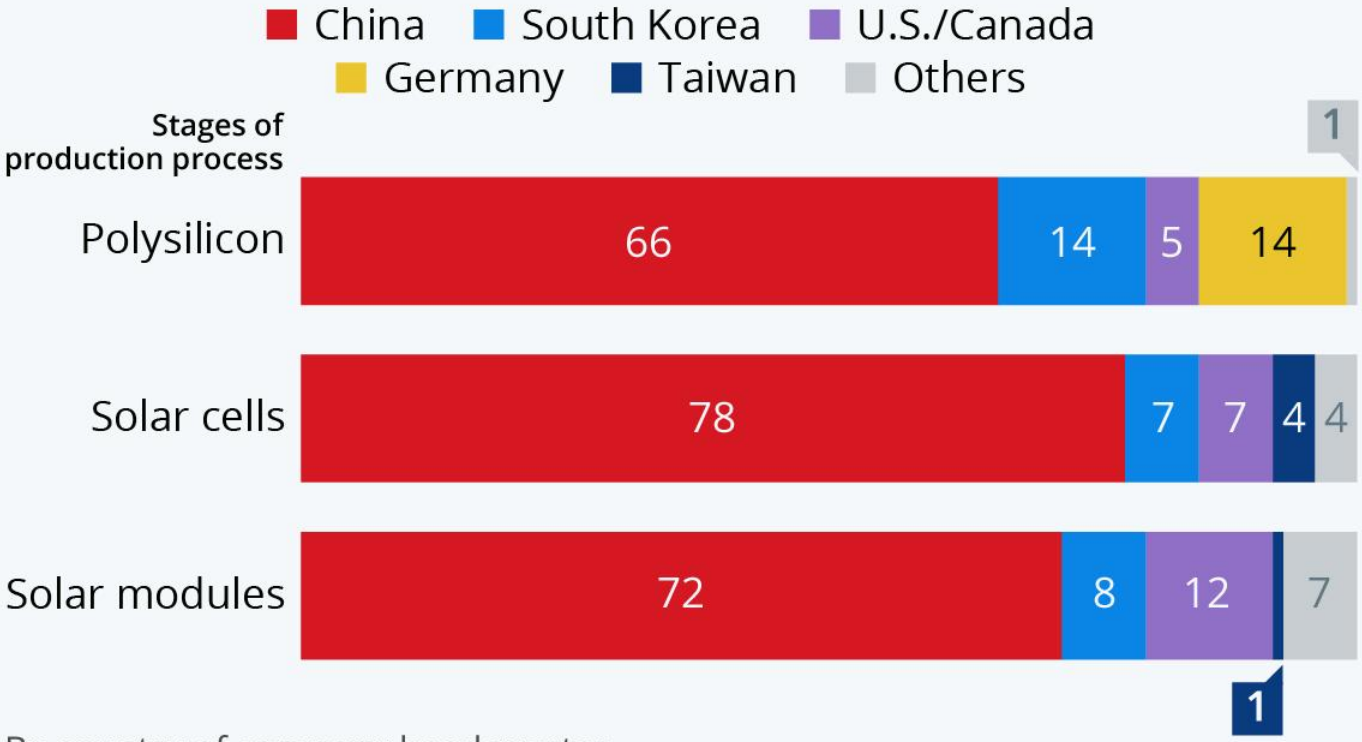
CC BY

Clean energy has gotten cheaper and cheaper!

(The T in IPAT!)

China Dominates All Steps of Solar Panel Production

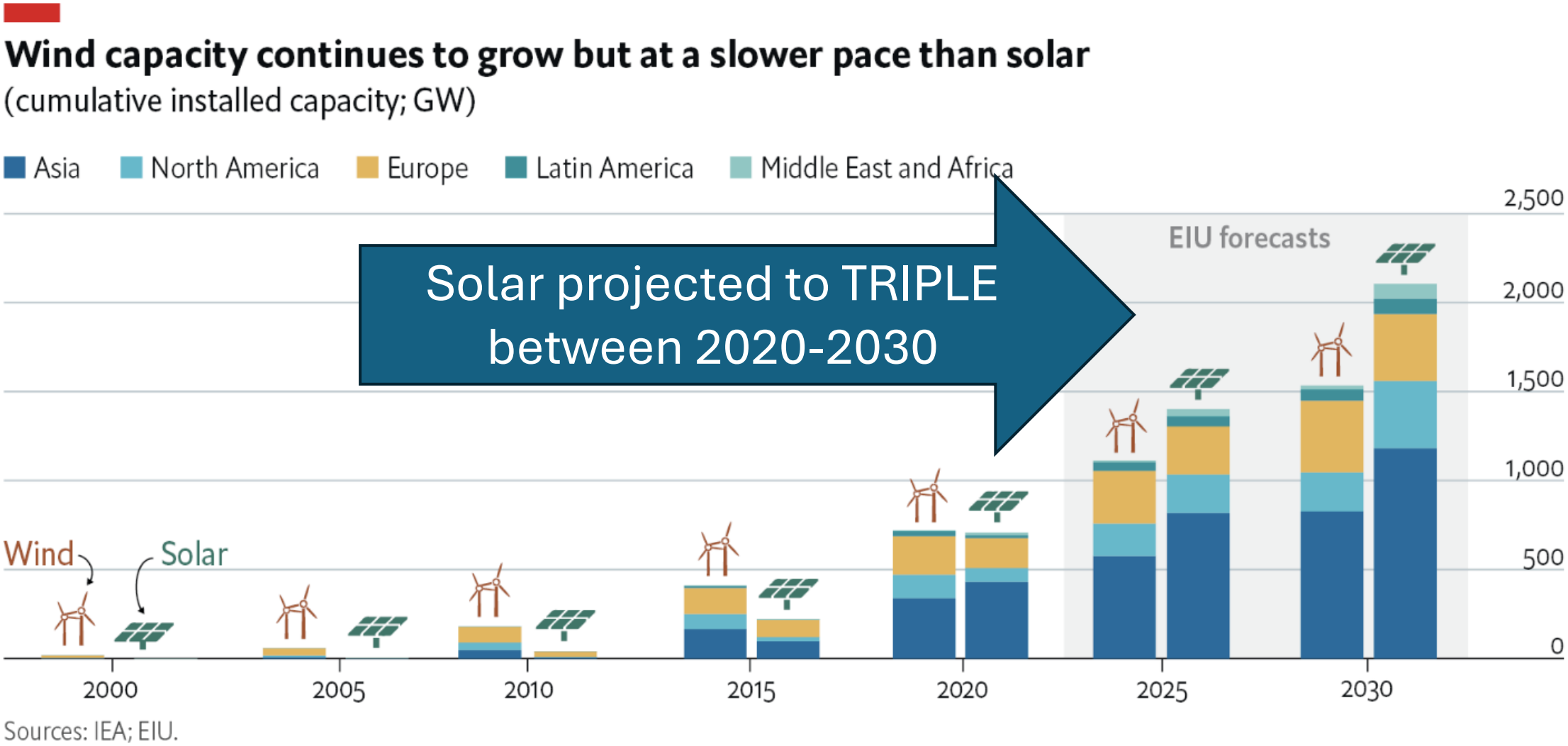
Country market shares of different products of global solar photovoltaic manufacturing in 2019 (in percent)



By country of company headquarter.
Totals measured in tons (polysilicon), gigawatts (cells, modules)
Source: Bloomberg NEF

statista

Clean Energy Growing Worldwide



Can Solar Eclipse Dirty Energy?

Climate Capital Renewable energy + Add to myFT

Chinese solar panel boom threatens Pakistan's debt-ridden grid

Industry rushes to switch to clean energy as cost of state power network becomes crippling

Solar panel plate at a market in Lahore. Business is racing to install cheap solar capacity © Arif Ali/AFP/Getty Images

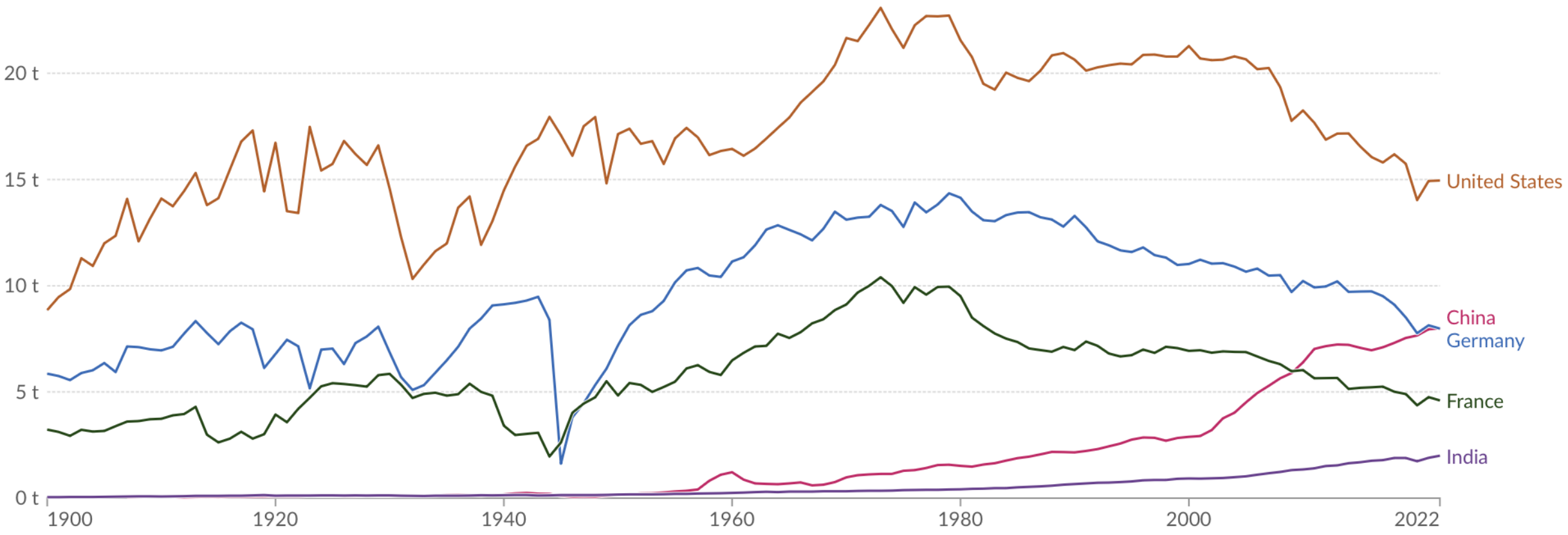
Per capita CO_2 emissions

Carbon dioxide (CO_2) emissions from fossil fuels and industry. Land-use change is not included.

Our World in Data

Table | Map | Chart

Settings



Play time-lapse 1800 2022

Data source: Global Carbon Budget (2023); Population based on various sources (2023) – Learn more about this data

OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

Annual CO_2 emissions

Carbon dioxide (CO_2) emissions from fossil fuels and industry. Land-use change is not included.

Our World in Data

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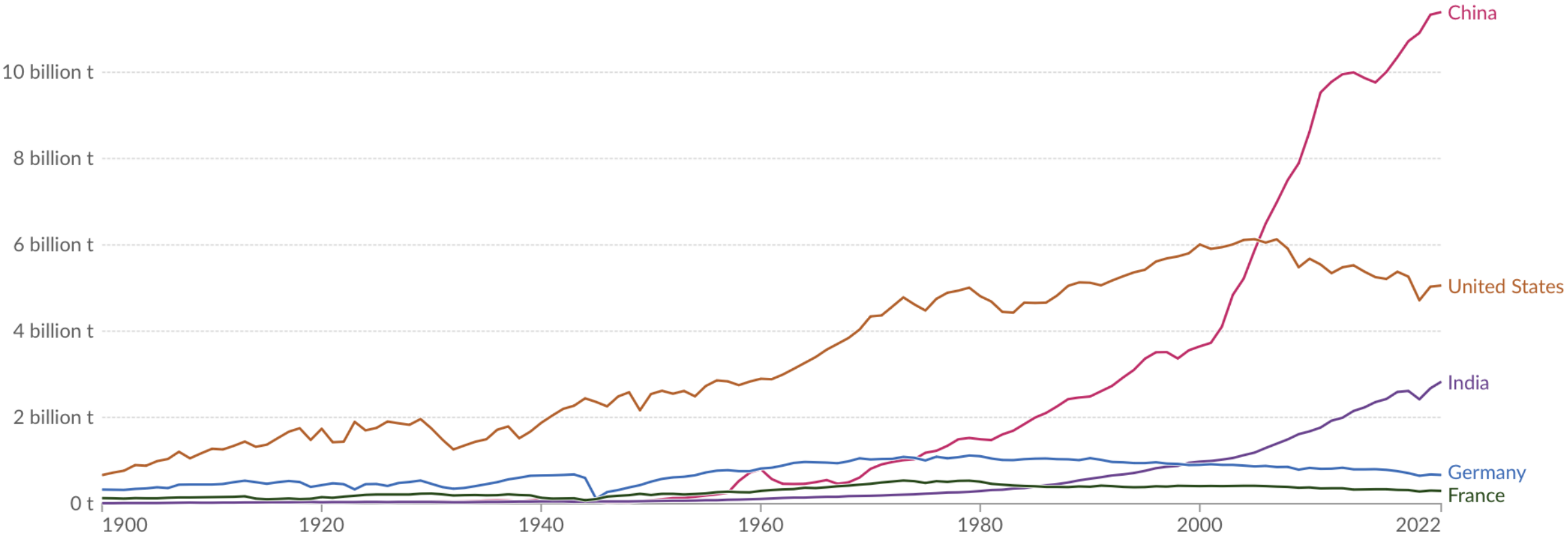
Settings



Play time-lapse 1792 2022

Data source: Global Carbon Budget (2023) – Learn more about this data

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Annual CO_2 emissions

Carbon dioxide (CO_2) emissions from fossil fuels and industry. Land-use change is not included.

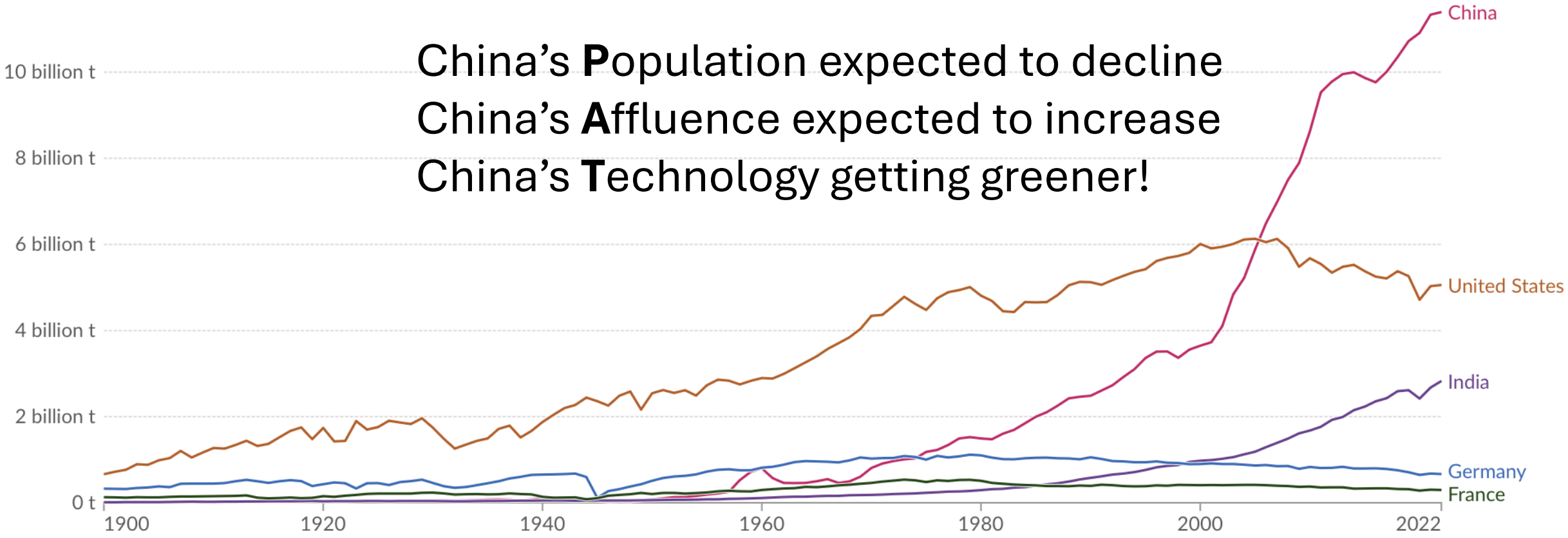
Our World in Data

Table | Map | Chart

Remember IPAT!

Settings

China's **P**opulation expected to decline
China's **A**ffluence expected to increase
China's **T**echnology getting greener!



Play time-lapse 1792 2022

Data source: Global Carbon Budget (2023) – Learn more about this data

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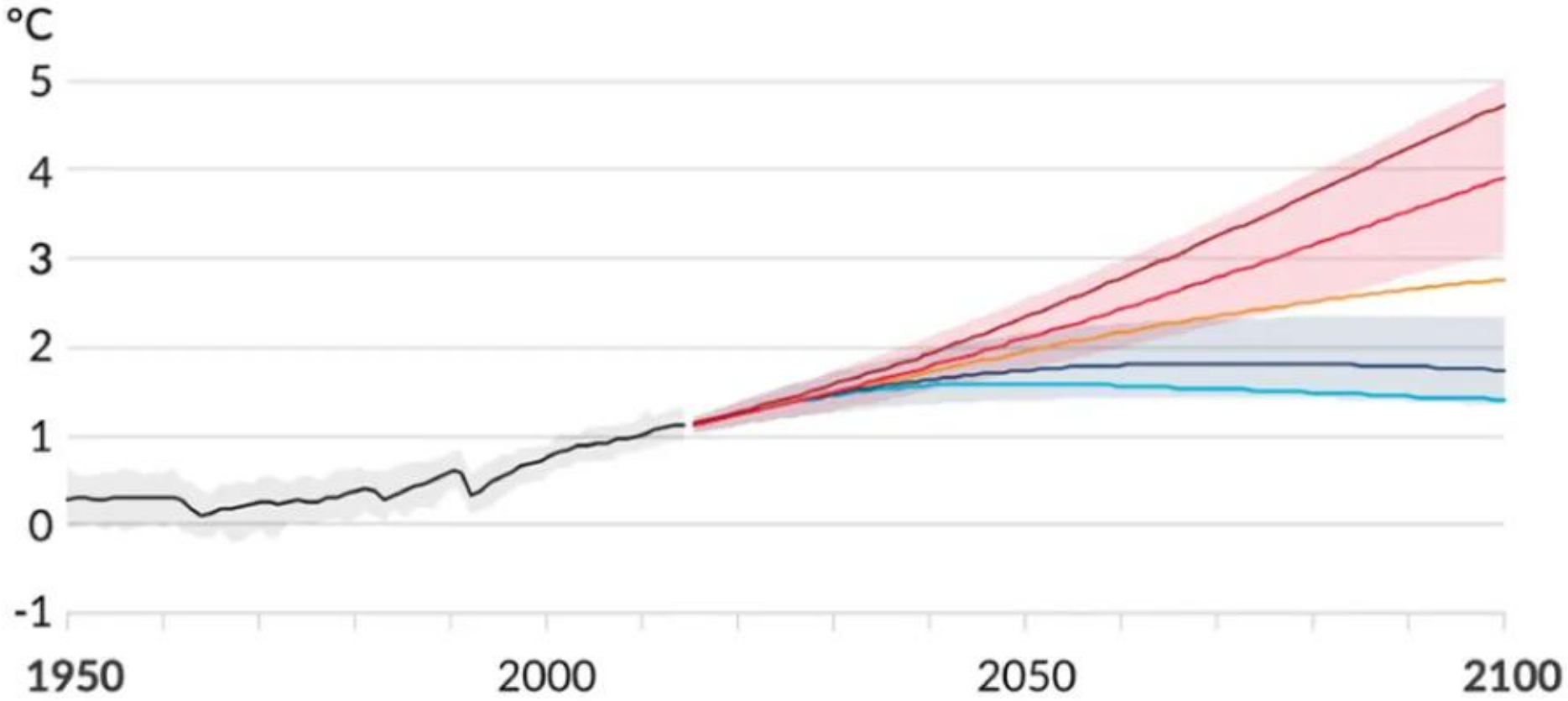
Projected Temperature Increase (°C)



Five Scenarios of Fossil Fuel Burning

- Highest CO_2 amounts
- Medium to high CO_2 amounts
- Medium CO_2 amounts
- Smaller CO_2 amounts, then no increase in CO_2 late in the 21st century
- No increase in CO_2 beginning in 2050

Consequences of Climate Change for Population

Fertility	Mortality	Migration

Caveat: Can’t rule out FUTURE dramatic changes. Lots of unknowns!

Consequences of Climate Change for Population

Fertility	Mortality	Migration
Very small effects *some people (esp. in Europe) report wanting fewer children*		

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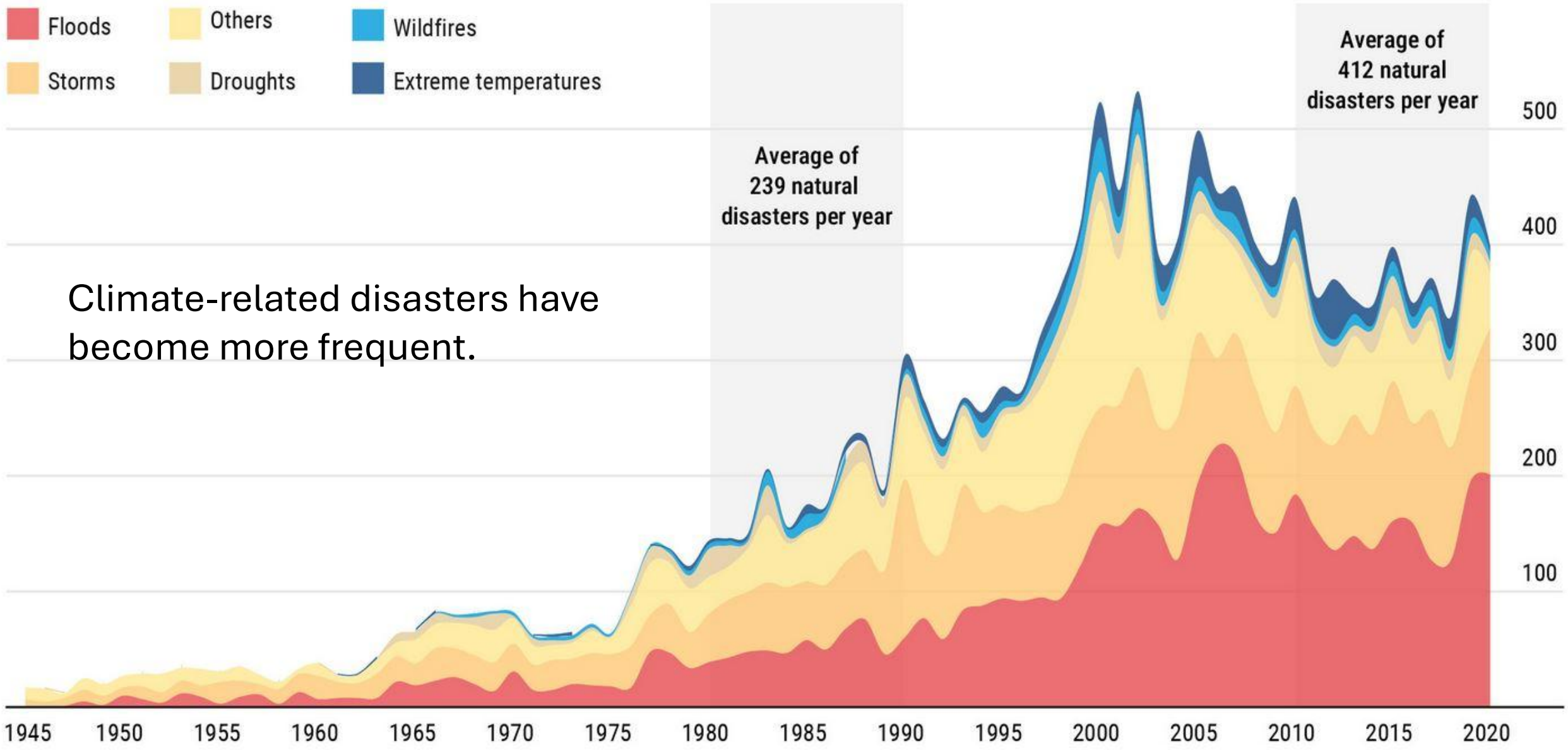
Consequences of Climate Change for Population

Fertility	Mortality	Migration
Very small effects *some people (esp. in Europe) report wanting fewer children*	Small relative to global population, but important impacts from disasters	

Caveat: Can't rule out FUTURE dramatic changes. Lots of unknowns!



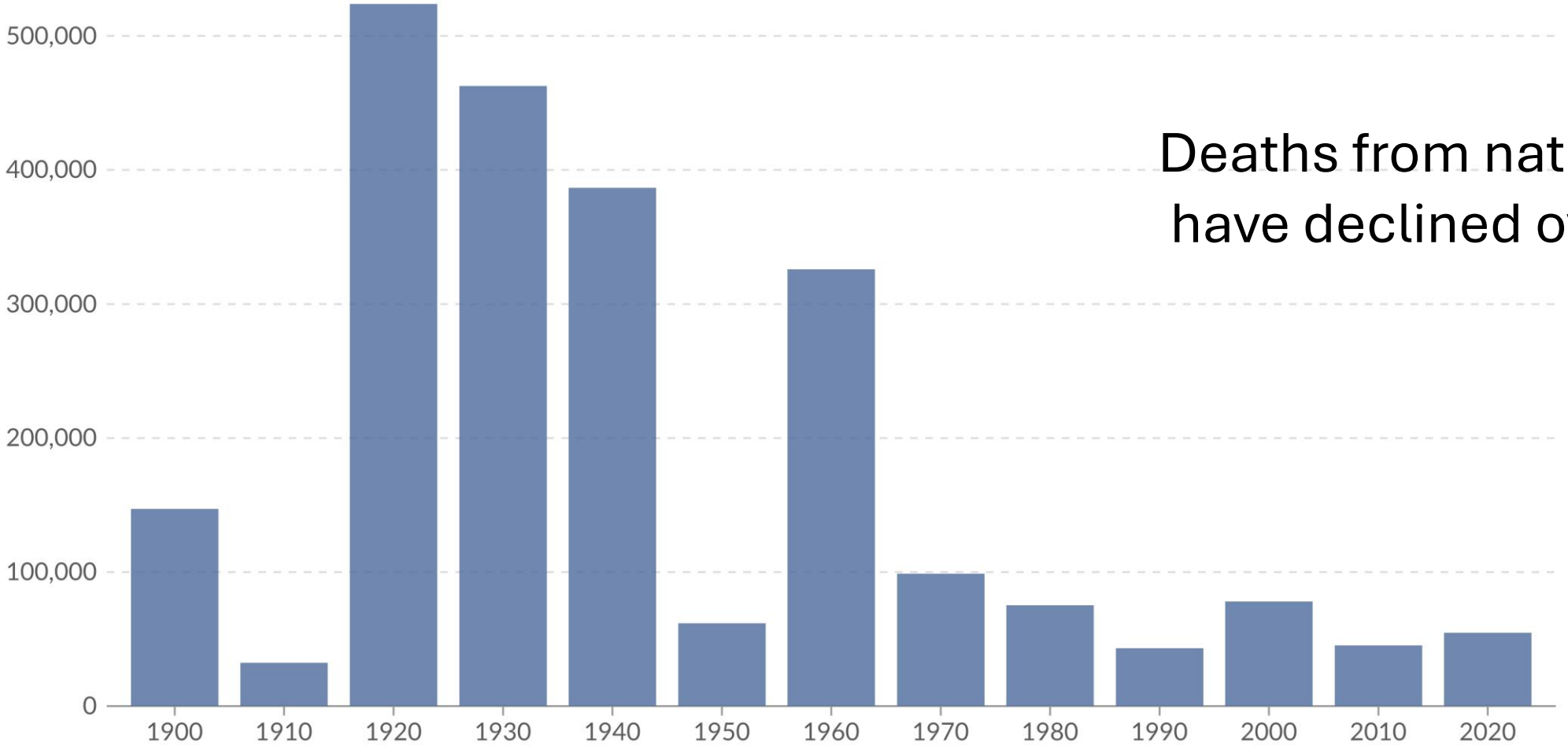
Climate-related disasters have become more frequent.

Source: World Trade Organization/Center for Research on the Epidemiology of Disasters (WTO/CRED).

Decadal average: Annual number of deaths from disasters, World

Our World in Data

Disasters include all geophysical, meteorological and climate events including earthquakes, volcanic activity, landslides, drought, wildfires, storms, and flooding. Decadal figures are measured as the annual average over the subsequent ten-year period.



Deaths from natural disasters have declined over time

Data source: EM-DAT, CRED / UCLouvain

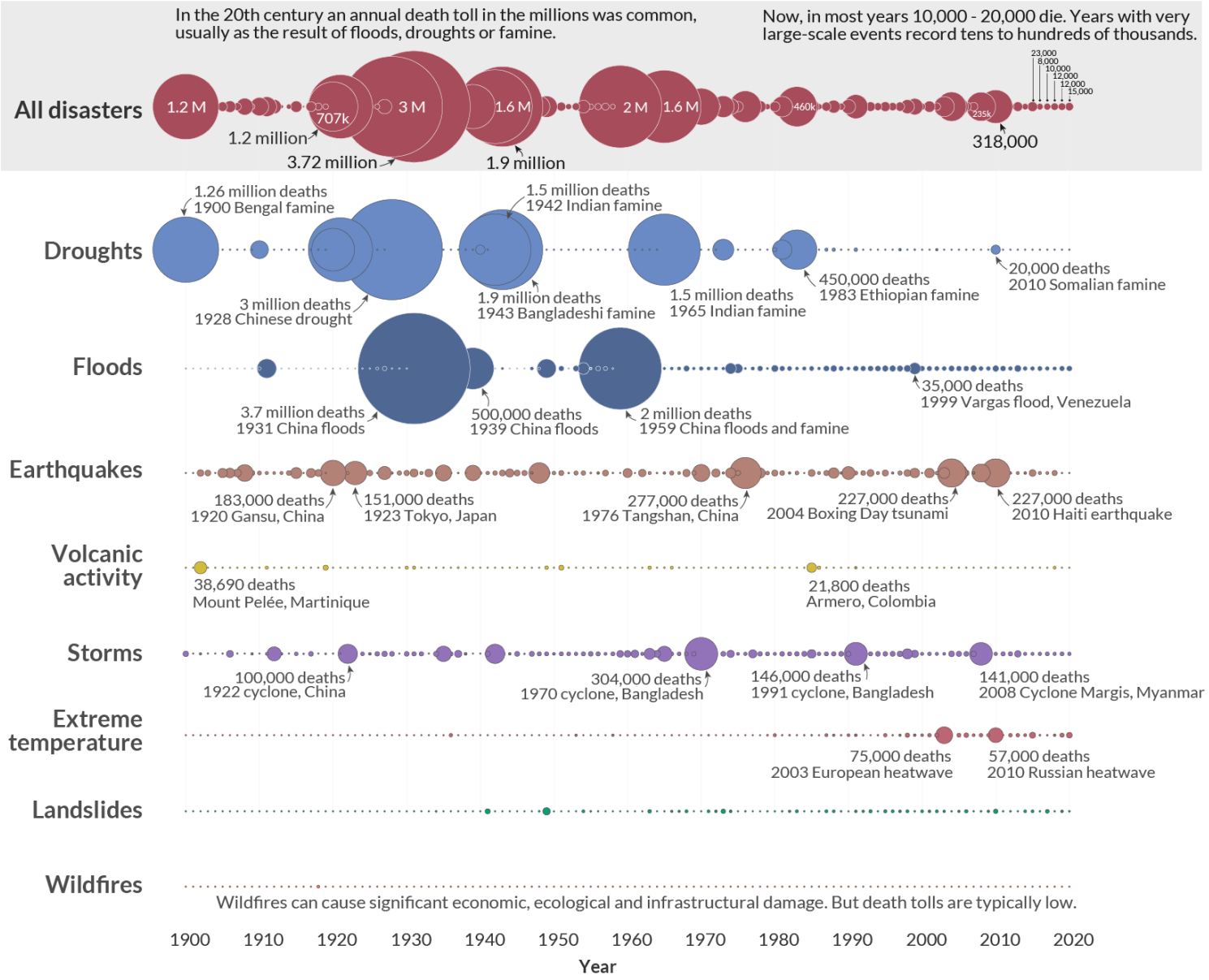
OurWorldinData.org/natural-disasters | CC BY

Note: Decadal figures are measured as the annual average over the subsequent ten-year period. This means figures for '1900' represent the average from 1900 to 1909; '1910' is the average from 1910 to 1919 etc. Data includes disasters recorded up to October 2025.

Global deaths from disasters over more than a century

The size of the bubble represents the estimated annual death toll. The largest years are labeled with this total figure, alongside large-scale events that contributed to the majority – although usually not all – of these deaths.

Our World in Data



Data source: EM-DAT, CRED / UCLouvain, Brussels, Belgium – www.emdat.be (D. Guha-Sapir).
OurWorldinData.org – Research and data to make progress against the world's largest problems.

Licensed under CC-BY by the author Hannah Ritchie.

Really big disasters (>1million deaths) are a thing of the past.

Consequences of Climate Change for Population

Fertility	Mortality	Migration
Very small effects *some people (esp. in Europe) report wanting fewer children*	Small relative to global population, but important impacts from disasters	**Big impacts.**

Caveat: Can't rule out FUTURE dramatic changes. Lots of unknowns!

Climate-driven migration: 2 causes

- *Immediate*: Displacement from natural disasters
- *Mitigating:* Responses to livelihood loss (and potential loss)

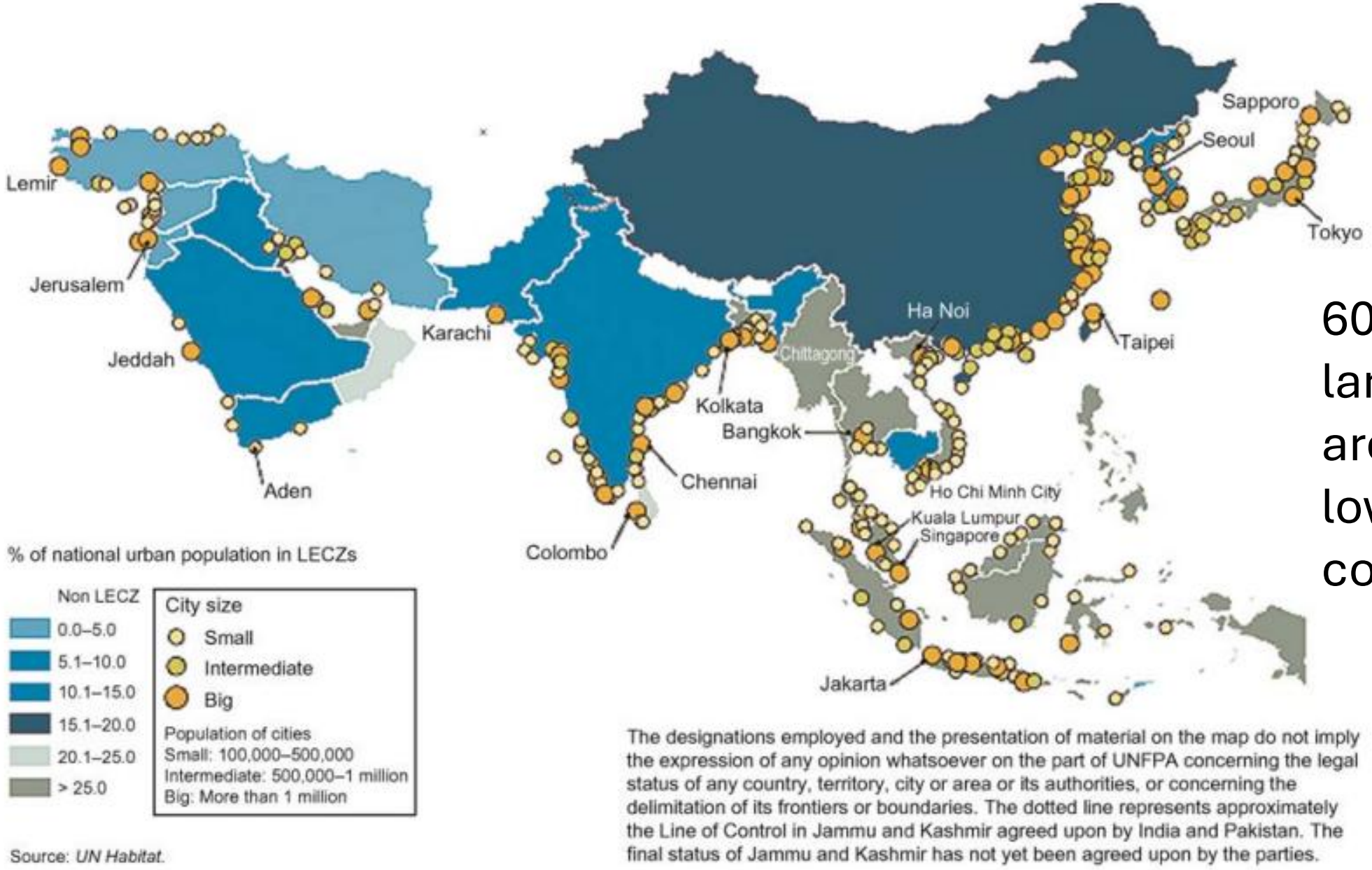
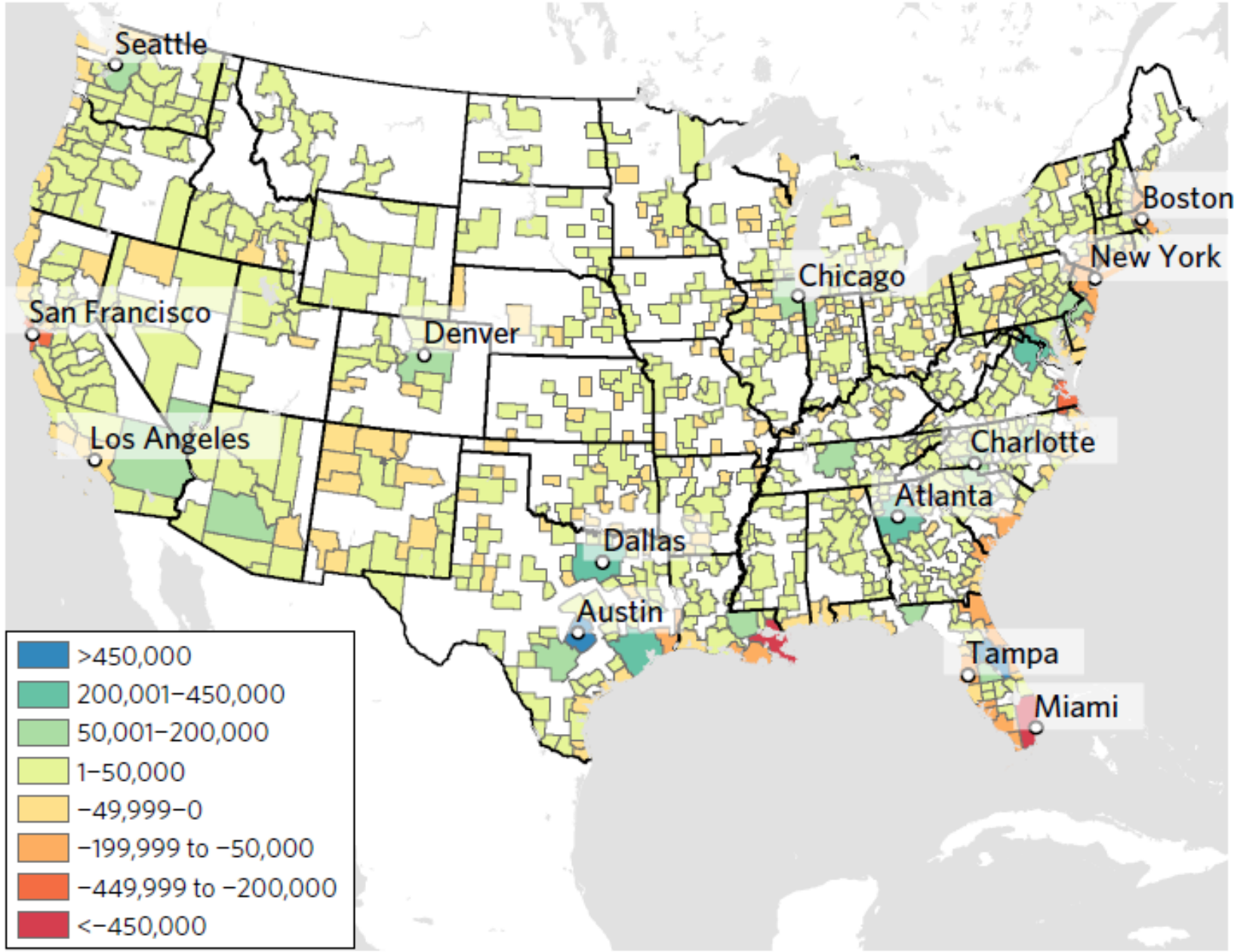


Fig. 4 Map of cities at low-elevation coastal zones (LECZ).

60% of world's largest cities are located in low-lying coastal areas



In the United States, 13.1 million people are at risk of migrating if there is a 1.8 m sea level rise by 2100

Lawrence Mower
@lmower3

NEW: Farmers Insurance says it's leaving Florida's troubled market.

Tens of thousands of Floridians could lose coverage.

w/ @harrisalexc:

tampabay.com
Farmers Insurance is leaving Florida in latest blow to homeowners
The decision was “necessary to effectively manage risk exposure,” the company said.

10:18 AM · Jul 11, 2023 · **181.6K** Views

145 753 1,183 24

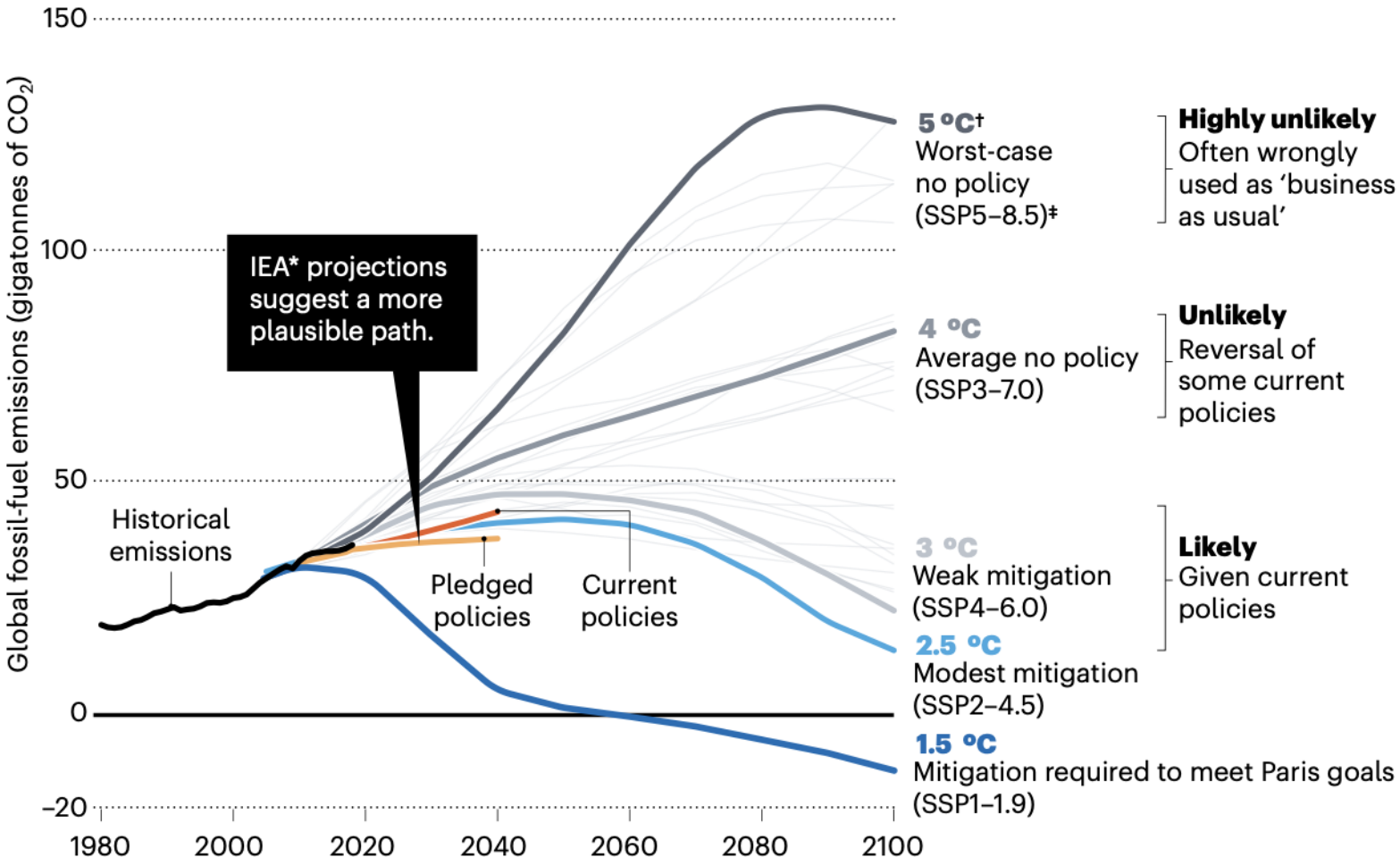
California insurance market rattled by withdrawal of major companies

Conclusions

- Population is only part of the equation (IPAT) affecting the environment
  - Not necessarily even the most important!
- But environmental change can have big impacts on population – particularly on migration

POSSIBLE FUTURES

The Intergovernmental Panel on Climate Change (IPCC) uses scenarios called pathways to explore possible changes in future energy use, greenhouse-gas emissions and temperature. These depend on which policies are enacted, where and when. In the upcoming IPCC Sixth Assessment Report, the new pathways (SSPs) must not be misused as previous pathways (RCPs) were. Business-as-usual emissions are unlikely to result in the worst-case scenario. More-plausible trajectories make better baselines for the huge policy push needed to keep global temperature rise below 1.5 °C.



*The International Energy Agency (IEA) maps out different energy-policy and investment choices. Estimated emissions are shown for its Current Policies Scenario and for its Stated Policies Scenario (includes countries' current policy pledges and targets). To be comparable with scenarios for the Shared Socioeconomic Pathways (SSPs), IEA scenarios were modified to include constant non-fossil-fuel emissions from industry in 2018.
†Approximate global mean temperature rise by 2100 relative to pre-industrial levels.
‡SSP5–8.5 replaces Representative Concentration Pathway (RCP) 8.5.

Question of the day

Thinking about what you have learned about the IPAT equation, population trends, and technological shifts, what do you think is the *easiest* way to reduce humanity's environmental impact? Explain in 1 sentence.

https://tinyurl.com/0205soc170